```
from google.colab import files
uploaded = files.upload()
```

اختيار الملفات 1) iris.csv
1) iris.csv(text/csv) - 3858 bytes, last modified: 13100% - 2026/4/ done
Saving 1) iris.csv to 1) iris.csv

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
df = pd.read_csv(r'1) iris.csv')
```

```
df.head(10)
```

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa
5	5.4	3.9	1.7	0.4	setosa
6	4.6	3.4	1.4	0.3	setosa
7	5.0	3.4	1.5	0.2	setosa
8	4.4	2.9	1.4	0.2	setosa
9	4.9	3.1	1.5	0.1	setosa

Next steps: Generate code with df New interactive sheet

```
df.tail(5)
```

	sepal_length	sepal_width	petal_length	petal_width	species
145	6.7	3.0	5.2	2.3	virginica
146	6.3	2.5	5.0	1.9	virginica
147	6.5	3.0	5.2	2.0	virginica
148	6.2	3.4	5.4	2.3	virginica
149	5.9	3.0	5.1	1.8	virginica

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   sepal_length  150 non-null    float64
 1   sepal_width   150 non-null    float64
 2   petal_length  150 non-null    float64
 3   petal_width   150 non-null    float64
 4   species       150 non-null    object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

```
df.describe()
```

	sepal_length	sepal_width	petal_length	petal_width
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.054000	3.758667	1.198667
std	0.828066	0.433594	1.764420	0.763161
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000

```
df.isnull().sum()
```

	0
sepal_length	0
sepal_width	0
petal_length	0
petal_width	0
species	0

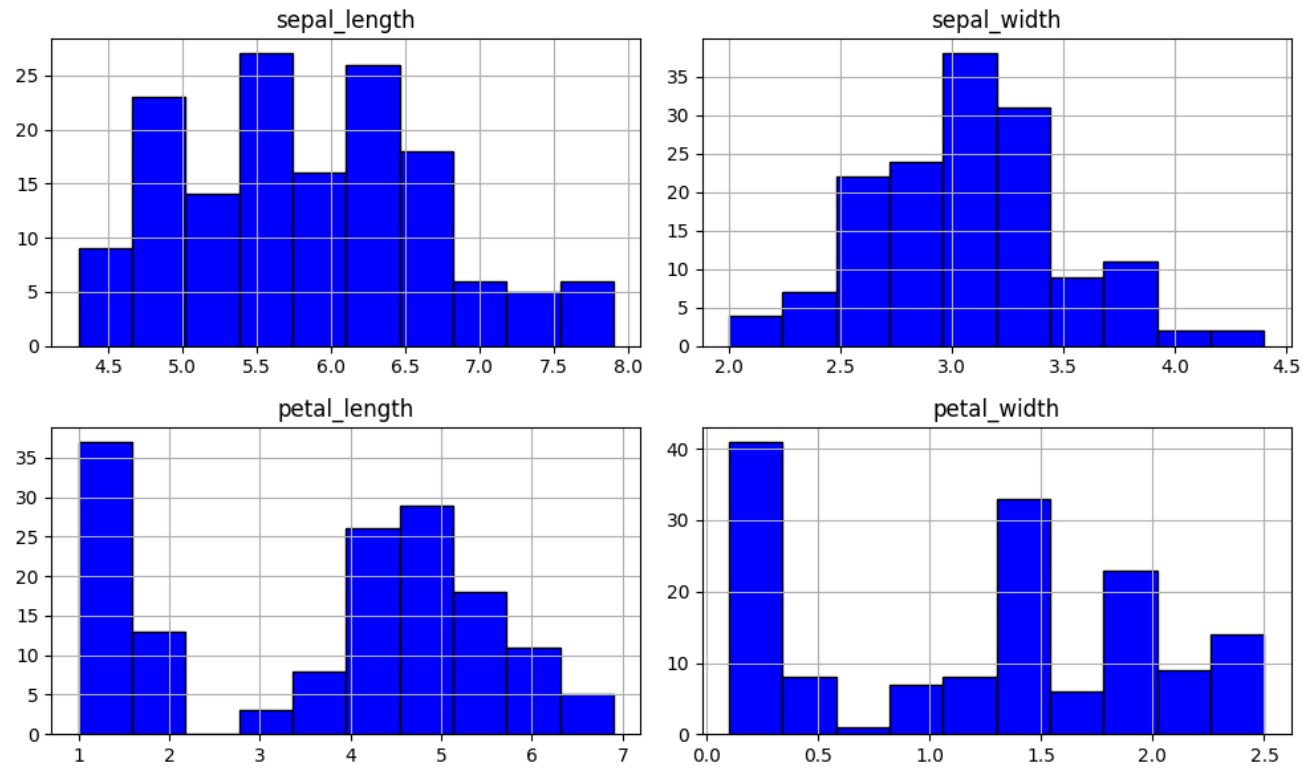
dtype: int64

```
df['species'].value_counts()
```

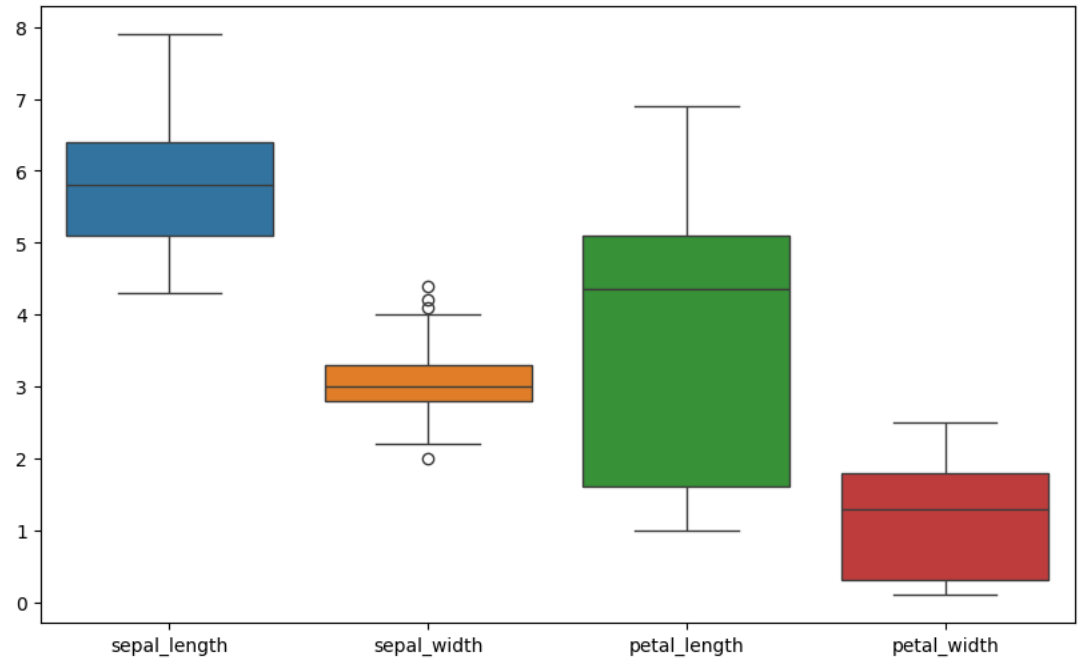
species	count
setosa	50
versicolor	50
virginica	50

dtype: int64

```
df.hist(color="blue",edgecolor="black",figsize=(10,6))
plt.tight_layout()
plt.show()
```

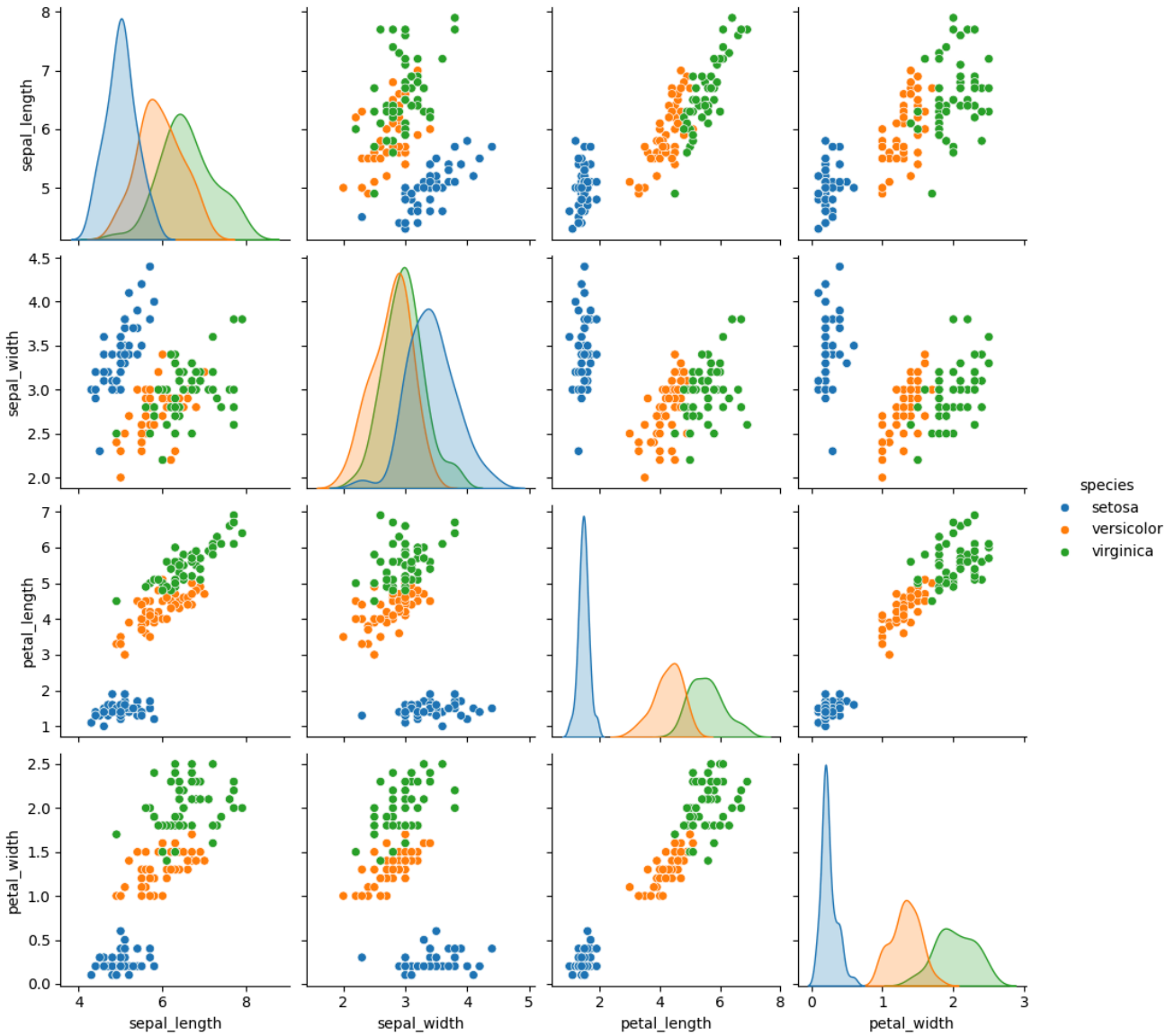



```
plt.figure(figsize=(10,6))
sns.boxplot(data=df)
plt.show()
```

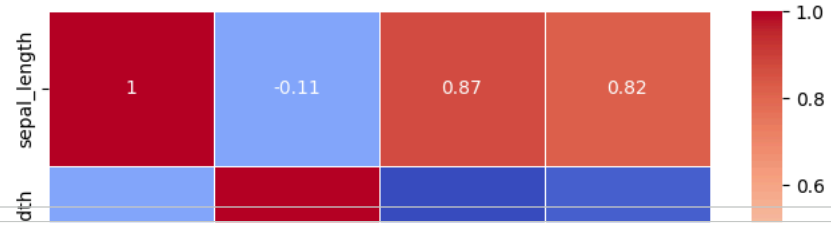



```
plt.figure(figsize=(10,6))
sns.pairplot(df,hue="species")
plt.show()
```

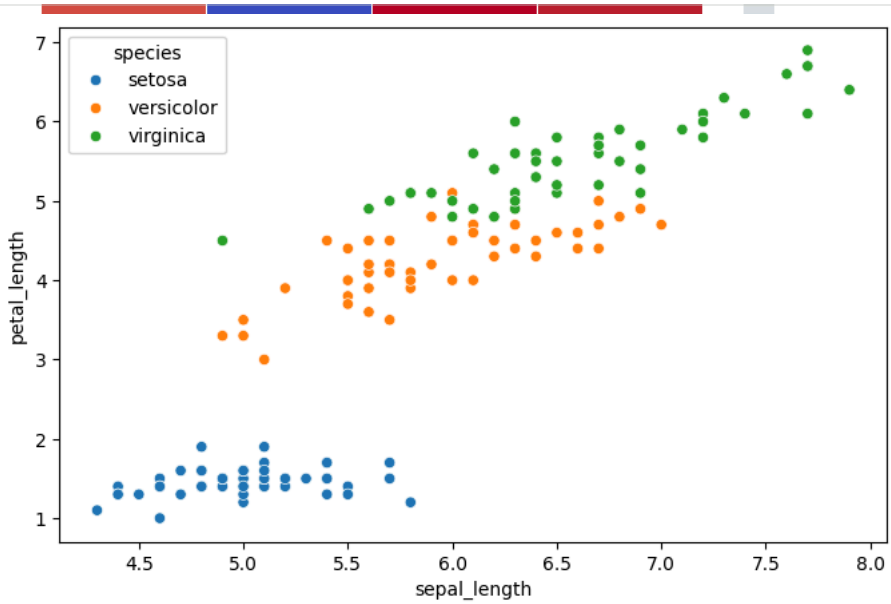
```
<Figure size 1000x600 with 0 Axes>
```




```
plt.figure(figsize=(8, 6))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
plt.show()
```




```
plt.figure(figsize=(8, 5))
sns.scatterplot(data=df,x='sepal_length', y='petal_length',hue='species')
plt.show()
```




```
plt.pie(df['species'].value_counts(),labels=df['species'].unique(),autopct='%1.1f%%')
plt.show()
```